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Question 3 → Pseudocode

```
Start
Print " Enter number of sensors: "
Read sensors
Print " Enter number of hourly readings: "
Read hours

Declare 2D array temp [sensors] [hours]
Call inputReadings (sensors, hours, temp)

Print " --- Individual Sensor Reports --- "
For i = 0 to sensors - 1
    CALL analyseSensor(hours, temp[i], total, avg, max_t, min_t,
                       critical, warning, max_streak)
    Print "Sensor" + (i + 1) + ": Total = " + total + " Avg = " + avg + " Max:
        + max_t + " Min = " + min_t
    Print "  Critical = " + critical + " Warning = " + warning + " Longest
        hot streak = " + max_streak
End For

best_sensor = highestAverageSensor (sensors, hours, temp)
best_total = 0
For j = 0 to hours - 1
    best_total += temp [best_sensor] [j]
End For
best_avg = best_total / hours
Print " Highest average: Sensor " + (best_sensor + 1) + " (" + best_avg
END
```


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//Function to input readings into 2D array
Function inputReadings (sensors, hours, temp)
    For i = 0 to sensors - 1
        Print "Enter readings for Sensor" + (i + 1) + ":"
        For j = 0 to hours - 1
            Read temp[i][j]
        End for
    End for
End function

//Function to analyze a single sensor's readings
Function analyzeSensor (hours, row_data, total, avg, max_t
    , min_t, critical, waring, max_streak)
total = 0
max_t = row_data[0]
min_t = row_data[0]
critical = 0
waring = 0
max_streak = 0
current_streak = 0

For j = 0 to hours - 1
    val = row_data[j]
    total = total + val
    If val > max_t then max_t = val End If
    If val < min_t then min_t = val End If
    If val >= 40 then
        critical += 1
    End If
    If val >= 35 AND val <= 39 then
        waring += 1
    End If
```


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```
If val >= 35 then
    current_streak += 1
    If current_streak > max_streak then
        max_streak = current_streak
    End If
Else
    current_streak = 0
End If
End for
avg = total / hours
END function

// Function to find the sensor with the highest avg temp
Function highestAverageSensor(sensors, hours, temp)
    best_sensor = 0
    max_avg = -1.0
    For i = 0 to sensors - 1
        total = 0
        For j = 0 to hours - 1
            total = total + temp[i][j]
        End for
        avg = total / hours

        If i == 0 or avg > max_avg then
            max_avg = avg
            best_sensor = i
        End If
    End for
    Return best_sensor
END function
```


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